

ADVANCED PC DISASSEMBLY AND COMPONENT ANALYSIS

1. Remove the side panel of a desktop PC case and take a clear photo showing the internal layout before any components are removed.

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2. Carefully remove the RAM module(s) from the motherboard, document the removal steps, and record the RAM type, speed, and form factor as printed on the module.

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Shut down the PC and disconnect the power cable.

Remove the side panel and locate the RAM slots on the motherboard.

Open the locking clips at both ends of the RAM slot.

Hold the RAM module **by its edges**—do not touch the gold contacts.

Gently lift the RAM module straight out of the slot. Take a clear photo of the **removed RAM module**, especially the label containing its specifications.

RAM details to record

- **Type:** DDR3 / DDR4 / DDR5
 - **Speed:** e.g., 2666 MHz / 3200 MHz / 4800 MT/s
 - **Form factor:** DIMM (desktop RAM)
3. Disconnect and remove the storage drive (HDD or SSD) from the system, then list all connector types (SATA, power, etc.) used and describe how the drive was mounted inside the case.
Hint: Pay attention to both cable connectors and any brackets or screws securing the drive.

-> Storage drive removal checklist

1. Identify the drive

- HDD or 2.5"/3.5" SATA SSD.
- Note the manufacturer/model if required.

2. Disconnect the cables

- **SATA data connector:** Usually a narrow, L-shaped 7-pin connector running from the drive to the motherboard.
- **SATA power connector:** Wider, L-shaped 15-pin connector coming from the power supply (PSU).
- Some systems may instead use an **M.2 drive**, which has **no SATA data or power cables**; it is secured directly to the motherboard with a small screw.

3. Check how the drive is mounted

- **3.5" HDD:** commonly mounted in a drive cage/tray using four screws, sometimes through rubber vibration-isolating mounts.
- **2.5" SSD:** commonly attached to a 2.5" mounting bracket/tray with four small screws.
- **Tool-less tray:** Some cases use plastic rails/clips instead of screws.
- **M.2 SSD:** inserted into an M.2 slot and held down with a small retaining screw.

4. Remove it

- Disconnect the connectors by holding the plugs rather than pulling the cables.
- Remove the retaining screws/clips or release the drive tray.
- Slide/lift the drive out carefully.

Example observation to record

Item	Observation
Drive type	3.5" SATA HDD
Data connector	SATA 7-pin
Power connector	SATA 15-pin
Mounting method	Drive cage
Fasteners	4 screws
Bracket/tray	None / metal drive tray

Important: The exact connector and mounting arrangement depends on the computer. If you upload a photo of the inside of the case and the drive, I can identify **each connector and mounting method specifically**.

4. Identify and remove one expansion card (such as a graphics card or Wi-Fi card), then document the connector type (e.g., PCIe), screw type used, and any locking mechanisms you encountered.

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Item	Observation
Expansion card	Graphics card (GPU)
Connector	PCIe x16
Slot	PCI Express x16 motherboard slot

Item	Observation
Screw	Phillips-head case/bracket screw
Locking mechanism	PCIe slot retention clip at the end of the slot
Additional connector	6-pin/8-pin PCIe power connector from the PSU, if required
Mounting	Metal rear bracket secured to the case

Removal procedure

1. Shut down the computer, unplug the power cable, and use appropriate **ESD protection**.
2. Open the case and locate the expansion card.
3. If present, disconnect the **6-pin/8-pin PCIe power cable** from the graphics card.
4. Remove the **Phillips-head screw(s)** securing the card's rear bracket to the case.
5. Locate the small **PCIe retention clip** at the end of the motherboard slot and release it.
6. Hold the card by its edges and gently pull it straight out of the PCIe slot.
7. Place the card on an **anti-static surface**.

Example lab-record answer

Expansion card removed: Graphics card

Connector type: PCIe x16

Screw type: Phillips-head screw securing the rear bracket

Locking mechanism: PCIe slot retention clip

Other connector: 8-pin PCIe power connector from the PSU

Mounting: The card was inserted into a PCIe x16 slot and its rear metal bracket was secured to the case with a screw.

**5. Explain three safety precautions you followed during the disassembly process to prevent static electricity damage or injury, and describe how you ensured proper grounding.

Hint: Mention tools or accessories like anti-static wrist straps, and steps like unplugging the power supply..**

-> Three Safety Precautions During Disassembly

1. Disconnected the power supply

- I shut down the computer completely and unplugged the power cable from the wall and the power supply before opening the case.
- This reduced the risk of electric shock and prevented damage from powered components.

2. Used an anti-static wrist strap

- I wore an **anti-static (ESD) wrist strap** and connected it to an appropriate grounded point on the computer chassis/workstation.
- This helped safely discharge static electricity from my body before touching components.

3. Handled components correctly

- I held RAM, expansion cards, and other components **by their edges**, avoiding direct contact with their pins, contacts, and circuitry.
- I placed removed components on an **anti-static surface/bag** rather than directly on a metal or conductive surface.